

CHEMISTRY

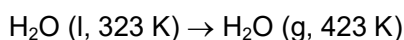
TARGET : JEE- Advanced

CAPS-20**THERMODYNAMICS****SINGLE CHOICE QUESTIONS**

1. A heating coil is immersed in a 100 g sample of $\text{H}_2\text{O}(\text{l})$ at 1 bar and 100°C in a closed vessel. In this heating process, 60% of the liquid is converted into gaseous form at constant pressure of 1 bar. Densities of liquid and gaseous water under these conditions are 1000 kg/m^3 and 0.60 kg/m^3 respectively. Magnitude of the work done for the process is:

(A) 4997 K (B) 4970 J (C) 9994 J (D) None of these

2. What is the value of change in internal energy at 1 atm in the process?



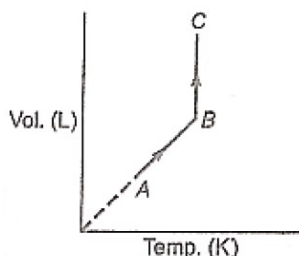
Given: $C_{v,m}(\text{H}_2\text{O}, \text{l}) = 75.0 \text{ JK}^{-1} \text{ mol}^{-1}$; $C_{p,m}(\text{H}_2\text{O}, \text{g}) = 33.314 \text{ JK}^{-1} \text{ mol}^{-1}$

ΔH_{vap} at 373 K = 40.7 kJ/mol

(A) 42.91 kJ/mol (B) 43086 kJ/mol (C) 42.6 kJ/mol (D) 49.6 kJ/mol

3. Two moles of a triatomic linear gas (neglect vibration degree of freedom) are taken through a reversible process ideal starting from A as shown in figure.

The volume ratio $\frac{V_B}{V_A} = 4$. If the temperature at A is -73°C , then:



Total enthalpies change in both steps is:

(A) 3000 R (B) 4200 R (C) 2100 R (D) 0

4. For polytropic process $PV^n = \text{constant}$, molar heat capacity (C_m) of an ideal gas is given by

(A) $C_{v,m} + \frac{R}{(n-1)}$ (B) $C_{v,m} + \frac{R}{(1-n)}$ (C) $C_{v,m} + R$ (D) $C_{p,m} + R \frac{R}{(n-1)}$

5. One mole of an ideal monoatomic gas at 27°C is subjected to a reversible iso-entropic compression until final temperature reaches to 327°C . If the initial pressure as 1.0 atm then find the value of $(\ln P_2)$: (Given: $\ln 2 = 0.7$)

(A) 1.75 atm (B) 0.176 atm (C) 1.0395 atm (D) 2.0 atm



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6. Using listed information's, calculate $\Delta_r G^\circ$ (in kJ/mol) at 27°C

	$\text{Co}_3\text{O}_4(\text{s}) +$	$4\text{CO}(\text{g}) \rightarrow$	$3\text{Co}(\text{s}) +$	$4\text{CO}_2(\text{g})$
Given: At 300 K	$\Delta_f H^\circ$ (kJ/mol) -891,	-110.5,	0.0,	-393.5
	S° (J/K-mol) 102.5,	197.7,	30.0,	213.7
(A) -214.8	(B) -195.0	(C) -200.3	(D) -256.45	

7. Consider the following data:

$$\Delta_f H^\circ (\text{N}_2\text{H}_4, \text{l}) = 50 \text{ kJ/mol}, \Delta_f H^\circ (\text{NH}_3, \text{g}) = -46 \text{ kJ/mol}$$

$$\text{B.E. (N-H)} = 393 \text{ kJ/mol and B.E. (H-H)} = 436 \text{ kJ/mol}$$

$$\Delta_{\text{vap}} H (\text{N}_2\text{H}_4, \text{l}) = 18 \text{ kJ/mol}$$

The N-N bond energy in N_2H_4 is:

- (A) 226 kJ/mol (B) 154 kJ/mol (C) 190 kJ/mol (D) None of these
8. If enthalpy of hydrogenation of $\text{C}_6\text{H}_6(\text{l})$ into $\text{C}_6\text{H}_{12}(\text{l})$ is -205 kJ/mol and resonance energy of $\text{C}_6\text{H}_6(\text{l})$ is -152 kJ/mol then enthalpy of hydrogenation of



is Assume ΔH_{vap} of $\text{C}_6\text{H}_6(\text{l})$, $\text{C}_6\text{H}_8(\text{l})$, $\text{C}_6\text{H}_{12}(\text{l})$ all are equal:

- (A) -535.5 kJ/mol (B) -238 kJ/mol (C) -357 kJ/mol (D) -119 kJ/mol
9. For the reaction taking place at certain temperature
- $$\text{NH}_2\text{COONH}_4(\text{s}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g}),$$
- if equilibrium pressure is $3X$ bar then $\Delta_r G^\circ$ would be
- (A) $-RT \ln 9 - 3RT \ln X$ (B) $RT \ln 4 - 3RT \ln X$
 (C) $-3RT \ln X$ (D) None of these

10. Gibbs Helmholtz equation relates the enthalpy, entropy and free energy change of the process at constant pressure and temperature as

$$\Delta G = \Delta H - T\Delta S \text{ (at constant P, T)}$$

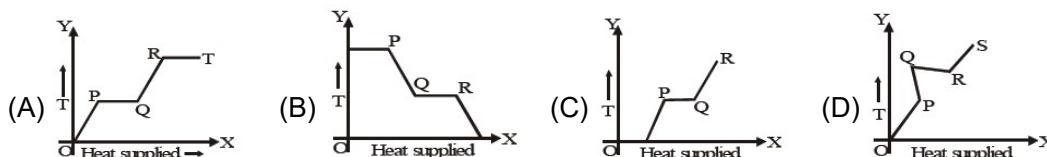
In general, the magnitude of ΔH does not change much with the change in temperature but the term $T\Delta S$ changes appreciably. Hence in some process spontaneity is very much dependent on temperature and such processes are generally known as entropy driven process.

For the reaction at 298 K ; $\text{A}_2\text{B}_4 \rightarrow 2\text{AB}_2$

$\Delta H = 2 \text{ kJ}$ and $\Delta S = 20 \text{ J/K}$ at constant P and T, the reaction will be

- (A) spontaneous and entropy driven (B) spontaneous and enthalpy driven
 (C) non-spontaneous (D) at equilibrium

11. Gibbs Helmholtz equation relates the enthalpy, entropy and free energy change of the process at constant pressure and temperature as
 $\Delta G = \Delta H - T\Delta S$ (at constant P, T)
 In general the magnitude of ΔH does not change much with the change in temperature but the term $T\Delta S$ changes appreciably. Hence in some process spontaneity is very much dependent on temperature and such processes are generally known as entropy driven process.
 When CaCO_3 is heated to a high temperature it decomposes into CaO and CO_2 , however it is quite stable at room temperature. It can be explained by the fact that
 (A) $\Delta_r H$ dominates the term $T\Delta S$ at high temperature
 (B) the term $T\Delta S$ dominates the $\Delta_r H$ becomes negative
 (C) at high temperature both $\Delta_r S$ and $\Delta_r H$ becomes negative
 (D) thermodynamics can not say anything about spontaneity
12. A block of ice at -10°C is slowly heated and converted into steam at 100°C . Which of the following curves represents the phenomenon qualitatively?



MULTIPLE CHOICE QUESTIONS

13. When ice melts at 1°C :
 (A) an increase in entropy
 (B) a decrease in enthalpy
 (C) a decrease in free energy
 (D) process is spontaneous
14. The value of $\Delta H_{\text{transition}}$ of $\text{C (graphite)} \rightarrow \text{C (diamond)}$ is 1.9 kJ/mol at 25°C . Entropy of graphite is higher than entropy of diamond. This implies that:
 (A) C (diamond) is more thermodynamically stable than C (graphite) at 25°C
 (B) C (graphite) is more thermodynamically stable than C (diamond) at 25°C
 (C) diamond will provide more heat on complete combustion at 25°C
 (D) $\Delta G_{\text{transition}}$ of $\text{C (diamond)} \rightarrow \text{C (graphite)}$ is $-\text{ve}$
15. For an isolated system, the entropy:
 (A) either increases or remains constant
 (B) either decreases or remains constant
 (C) can never decrease
 (D) can never increase

COMPREHENSION # 1 (FOR Q. 16 TO Q.20)

Standard Gibb's energy of reaction ($\Delta_r G^\circ$) at a certain temperature can be computed as $\Delta_r G^\circ = \Delta_r H^\circ - T \cdot \Delta_r S^\circ$ and the change in the value of $\Delta_r H^\circ$ and $\Delta_r S^\circ$ for a reaction with temperature can be computed as follows:

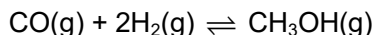
$$\Delta_r H^\circ_{T_2} - \Delta_r H^\circ_{T_1} = \Delta_r C_p^\circ (T_2 - T_1)$$

$$\Delta_r S^\circ_{T_2} - \Delta_r S^\circ_{T_1} = \Delta_r C_p^\circ \ln \left(\frac{T_2}{T_1} \right)$$

$$\Delta_r G^\circ = \Delta_r H^\circ - T \cdot \Delta_r S^\circ$$

$$\text{and by } \Delta_r G^\circ = -RT \ln K_{eq}.$$

Consider the following reaction:



$$\text{Given: } \Delta_f H^\circ (\text{CH}_3\text{OH, g}) = -201 \text{ kJ/mol; } \Delta_f H^\circ (\text{CO, g}) = -114 \text{ kJ/mol}$$

$$S^\circ (\text{CH}_3\text{OH, g}) = 240 \text{ J/mol-K; } S^\circ (\text{H}_2, \text{g}) = 29 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$S^\circ (\text{CO, g}) = 198 \text{ J/mol-K; } C_{p,m}^\circ (\text{H}_2) = 28.8 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$C_{p,m}^\circ (\text{CO}) = 29.4 \text{ J/mol-K; } C_{p,m}^\circ (\text{CH}_3\text{OH}) = 44 \text{ J/mol-K}$$

$$\text{and } \ln \left(\frac{320}{300} \right) = 0.06, \text{ all data at 300 K.}$$

16. $\Delta_r S^\circ$ at 300 K for the reaction is:
 (A) 152.6 J/K-mol (B) 181.6 J/K-mol (C) -16 J/K-mol (D) None of these
17. $\Delta_r H^\circ$ at 300 K for the reaction is
 (A) -87 kJ/mol (B) 87 kJ/mol (C) -315 kJ/mol (D) -288 kJ/mol
18. $\Delta_r S^\circ$ at 320 K is
 (A) 155.18 J/mol-K (B) 150.02 J/mol-K (C) 172 J/mol-K (D) None of these
19. $\Delta_r H^\circ$ at 320 K is
 (A) -288.86 kJ/mol (B) -289.1 kJ/mol (C) -87.86 kJ/mol (D) None of these
20. $\Delta_r G^\circ$ at 320 K is
 (A) -48295.2 kJ/mol (B) -240.85 kJ/mol (C) 240.85 kJ/mol (D) -81.91 kJ/mol

MATRIX TYPE QUESTIONS

21. Column I and Column II contains four entries each. Entries of Column I are to be matched with some entries of column II. One or more than one entries of Column I may have the matching with the same entries of Column II.

Column-I (Process)	Column-II (Entropy Change)
(A) Reversible isothermal compression of an ideal gas	(P) $\Delta S_{\text{System}} > 0$
(B) Isothermal free expansion ($P_{\text{ext}} = 0$) of an ideal gas	(Q) $\Delta S_{\text{System}} < 0$
(C) Reversible adiabatic expansion of an ideal gas	(R) $\Delta S_{\text{System}} = 0$
(D) Reversible ideal gas expansion	(S) Information insufficient

22. Match the following

Column-I	Column-II
(A) $(\Delta G_{\text{system}})_{T,P}$	(P) $nR \ln \left(\frac{V_2}{V_1} \right)$
(B) Work done in reversible isothermal ideal gas expansion	(Q) $nRT \ln \left(\frac{P_2}{P_1} \right)$
(C) ΔG for reversible isothermal expansion of an ideal gas	(R) $-nFE$
(D) ΔS_{gas} for isothermal expansion of an ideal gas	(S) $nR \ln \left(\frac{P_1}{P_2} \right)$

23. Match the following

Column-I	Column-II
(A) $\left(\frac{\partial U}{\partial T} \right)_V$	(P) C_P
(B) $\left(\frac{\partial H}{\partial T} \right)_P$	(Q) C_V
(C) $\left(\frac{\partial G}{\partial T} \right)_P$	(R) $-S$
(D) $\left(\frac{\partial G}{\partial P} \right)_T$	(S) V

SUBJECTIVE ANSWER TYPE

24. One mole of an ideal monoatomic gas is carried through the cycle of the given figure consisting of step A, B and C and involving state 1, 2 and 3. Fill in the blank space in the table given below assuming reversible steps.

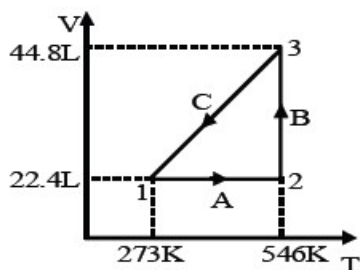


Table-1

State	P	V	T
1			
2			
3			

Step	Name of process	q	w	ΔE	ΔH
A					
B					
C					

ANSWER KEY CAPS-20

1. (C)	2. (C)	3. (B)	4. (B)	5. (A)
6. (D)	7. (C)	8. (D)	9. (D)	10. (A)
11. (B)	12. (A)	13. (ACD)	14. (BCD)	15. (AC)
16. (C)	17. (A)	18. (D)	19. (C)	20. (D)
21. (A)	22. (C)	23. (B)		